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CS370 Project 1 Encryption

3.1

plaintext : Im here!

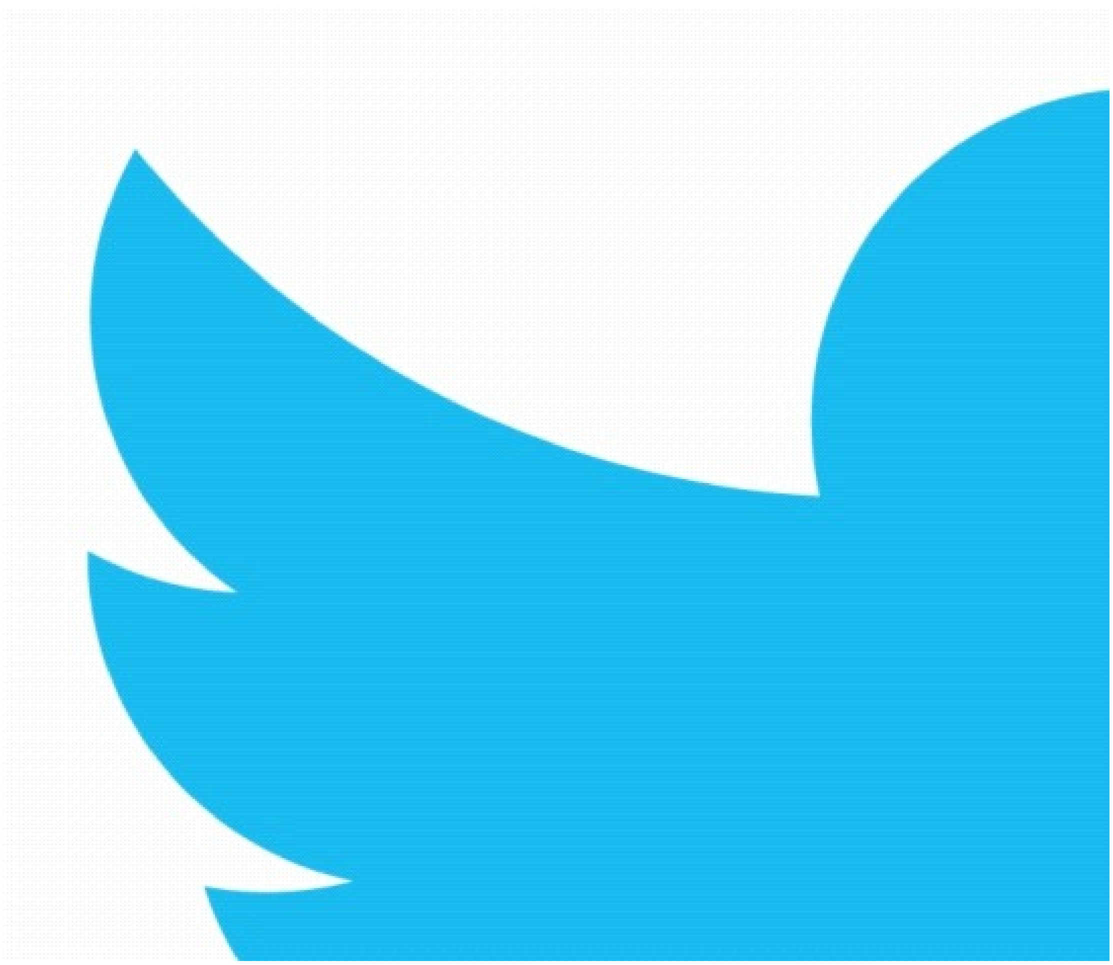
a. -aes-256-cbc password:123 chipertext: U2FsdGVkX188U0JFujHdR1bc3ecRnX45lRzkemsaP7k=

b. bf-cfb password 123 chiphertext: U2FsdGVkX191/F7kyu/4fQhCwNDn8Ikk

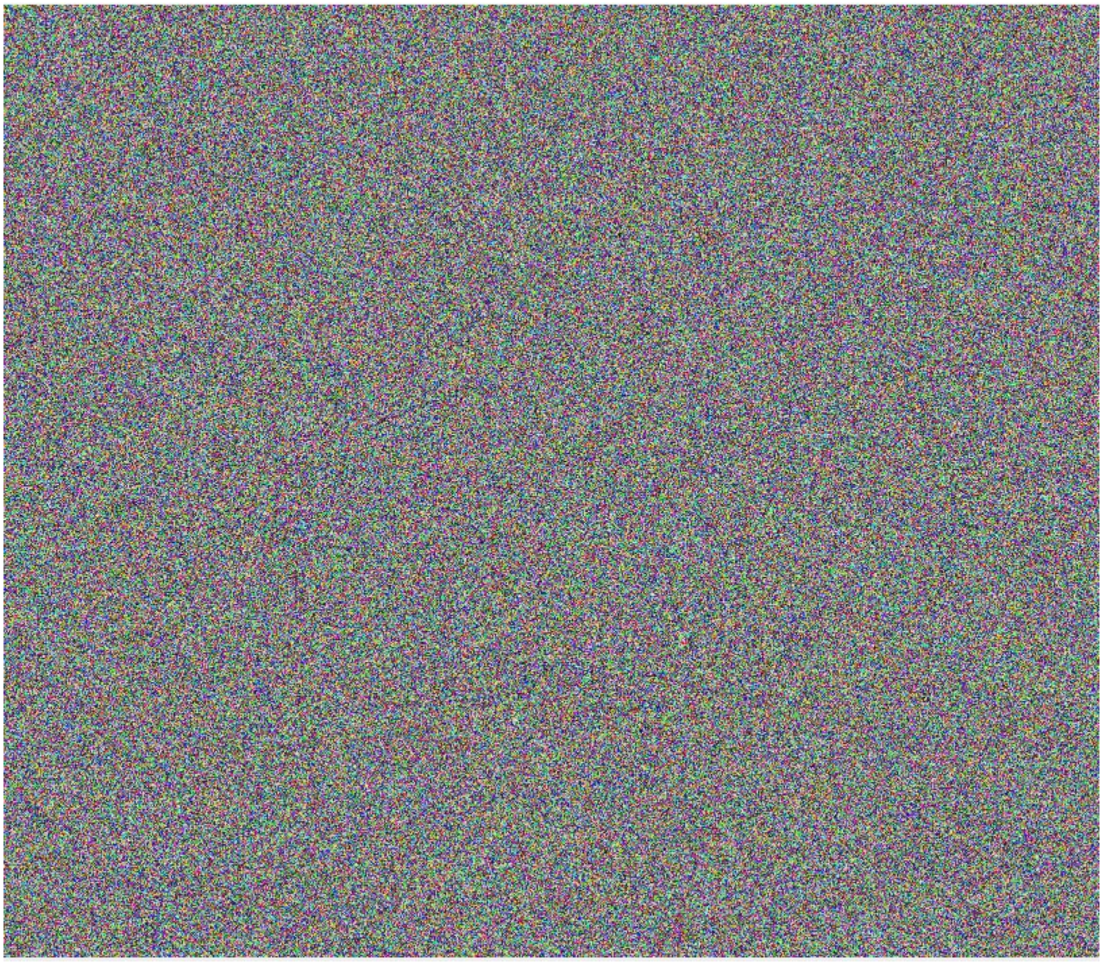
c. camellia-256-ecb password:123 chiphertext: U2FsdGVkX191/F7kyu/4fQhCwNDn8Ikk=

3.2

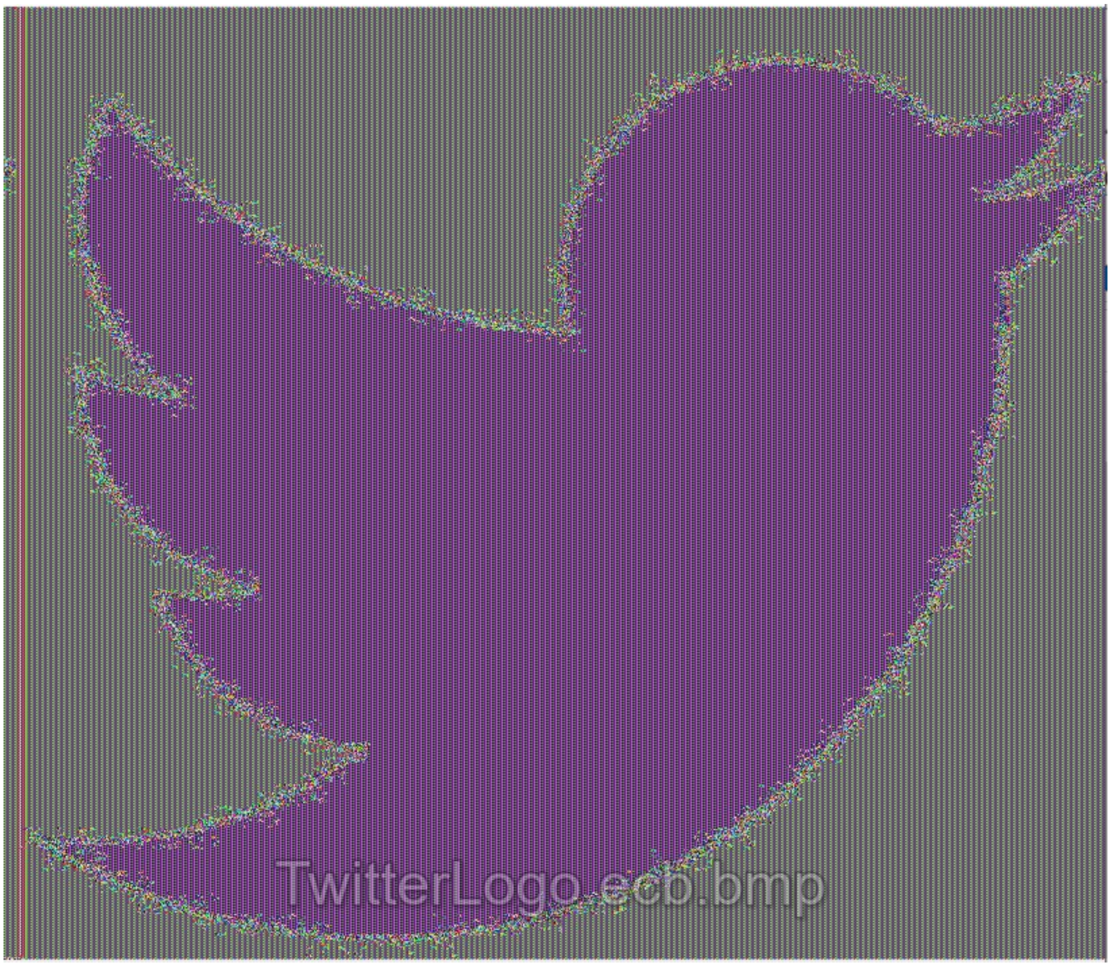
original



CBC



ECB



Here we can see patterns evolving because of the some of the weakness of ECB mode.

3.3

1 Encrypt1.bin and Encrypt2.bin did indeed have the same contents because it was run with the same key and same IV or the same starting the point. The IV acts as seed to start the encryption process so this generated the same cipher text. Encrypt1 and Encrypt3 did not have the same contents because the IV was different, in this case the seed or starting point was changed so the resulting output or cipher text was also different.

3.4 To use my openssl.py module use the function findKey(plain,cipher), so for the given example. findKey('8d20e5056a8d24d0462ce74e4904c1b513e10d1df4a2ef2ad4540fae1ca0aaf9','This is a top secret.')

3.5

SHA1(plain3.5.txt)= da39a3ee5e6b4b0d3255bfef95601890afd80709

SHA256(plain3.5.txt)= e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855

RIPEMD160(plain3.5.txt)= 9c1185a5c5e9fc54612808977ee8f548b2258d31

3.6

HMAC-SHA256(input3.6.txt)=
0612b53909144fd22afea0a785431327a657095ffea7d1e7f84d04fb96645726

Keys for HMAC can be any length becase they will first be hashed before being used.

3.7

Bit	SHA512	SHA256
1	1	1
49	0	4
73	0	4
113	0	4
149	1	5
173	1	5
1113	1	5
4973	0	8
49113	0	8
73113	0	8
14973	1	9
149113	1	9
173113	1	9
4973113	0	12

The pattern: When a single bit is flipped, there is only a one-bit change between H1 and H2 in SHA512 and SHA256. This is because flipping a single bit only changes a single bit in the input file, and one-way hash functions only examine the input file when computing the hash value. As a result, flipping a bit only affects a single bit in the file. When two bits are flipped, the difference between H1 and H2 in SHA256 is raised to four bits, and when two bits are flipped in SHA512, the difference is increased to five bits. If four bits were flipped the difference between SHA512 and SHA256 would be 12 bits.

3.8

It took many trials to find any success with either the strong or weak collision. On average the weak collision test took over 200 attempts to break and the strong took over 300 to break. Based on my observations week collision resistance is easier to break.